RAK11310 WisBlock LPWAN Module Datasheet

Overview

Description

RAK11310 is a **WisBlock Core** module for the RAK hardware IoT platform **WisBlock**. It is powered by an RP2040 ARM microcontroller developed by the Raspberry Pi Foundation, combined with the SX1262 LoRa transceiver from Semtech. **RAK11310** can be programmed via the Arduino™ IDE, PlatformIO or MicroPython.

The RAK11310 WisBlock Core module has a RAK11300 LoRa stamp module in it together with a high-quality connector that is compatible with WisBlock Base boards. It allows an easy way to access the pins of the RAK11310 module in order to simplify the development of IoT devices.

The module complies with LoRaWAN 1.0.2 protocols, and also supports LoRa point-to-point communication.

The RF communication characteristic of the Lora module makes it suitable for a variety of applications in the IoT field such as home automation, sensor networks, building automation, personal area networks applications (health/fitness sensors, and monitors, etc.).

Features

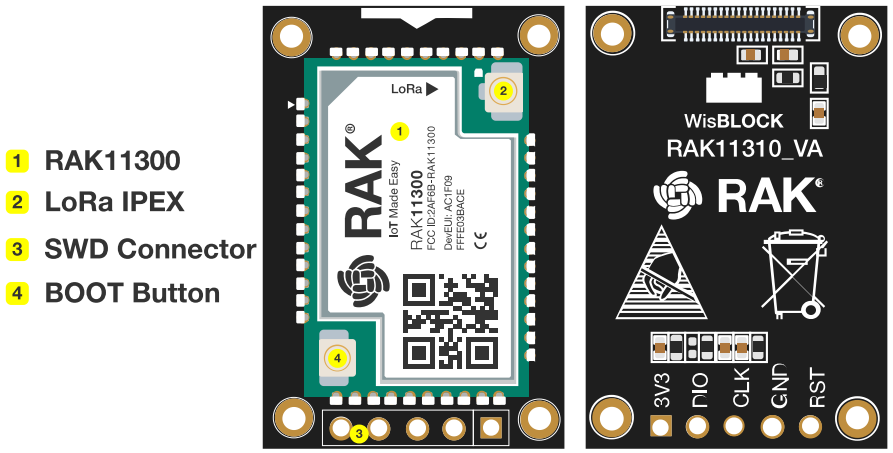
- Based on RAK11300
- Uses the RP2040 as the main processor
- Semtech SX1262 low power high range LoRa transceiver
- LoRaWAN 1.0.2 protocol stack
- **I/O ports**: UART/I2C/GPIO/USB
- Serial Wire Debug (SWD) interface
- **Module Size**: 20 x 30 mm
- **Supply Voltage**: 2.0 V ~ 3.6 V
- **Temperature Range**: -20 °C ~ 70 °C
- **Chipset**: Raspberry Pi Foundation RP2040, Semtech SX1262

Specifications

Overview

The overview covers the RAK11310 board overview and the mounting mechanics of the board into the baseboard.

Board Overview



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Figure 1: RAK11310 Overview

Mounting Sketch

The RAK11310 module is designed to work with the RAK5005-O base board. **Figure 2** shows how a RAK11310 module should be mounted on top of the RAK5005-O.

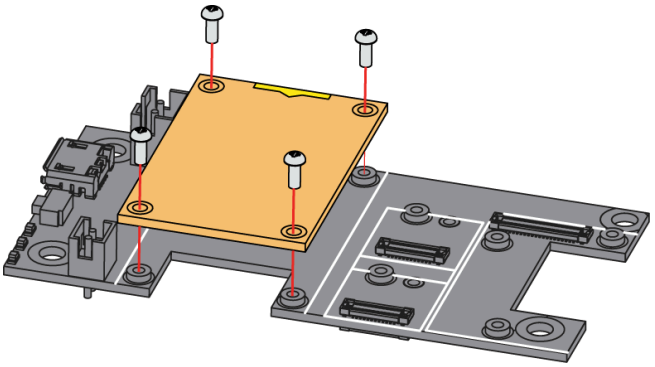


Figure 2: RAK11310 Mounting Sketch

Hardware

The hardware specification is categorized into four parts. It covers the RF, electrical, and mechanical parameters, which include the tabular data of the functionalities and standard values. It also shows and discusses the diagram of the RAK11310 WisBlock LPWAN Module.

Chipset

Vendor	Part number
Raspberry Pi Foundation, Semtech	RP2040, SX1262

Internal Connections Between RP2040 MCU and SX1262 LoRa transceiver

SX1262	RP2040	Function
SPI_NSS	GPIO13	SPI select
SPI_SCK	GPIO10	SPI clock
SPI_MISO	GPIO12	SPI Master in
SPI_MOSI	GPIO11	SPI Master out
NRESET	GPIO14	SX1262 reset
ANT_SW	GPIO25	SX1262 RF switch power
DIO1	GPIO29	SX1262 DIO1
BUSY	GPIO15	SX1262 BUSY

RF Characteristics

The RAK11310 module supports the LoRaWAN bands, as shown in the table below. There are two types RAK11310 module: **RAK11310(L)** is for the low-frequency band regions and **RAK11310** (no extra suffix) is for the high-frequency band regions.

NOTE

Check the frequency and band regions, as shown in the table, when ordering your RAK11310 WisBlock Core.

Module	Core Module	Region	Frequency
RAK11310(L)	RAK11300(L)	Europe	EU433
	RAK11300(L)	China	CN470
RAK11310	RAK11300	Europe	EU868
	RAK11300	North America	US915
	RAK11300	Australia	AU915
	RAK11300	Korea	KR920
	RAK11300	Asia	AS923
	RAK11300	India	IN865
	RAK11300	Russia	RU864

Electrical Characteristics

Power Consumption

NOTE

The power consumption will be published after the hardware test results.

Feature	Condition	Minimum	Typical	Maximum	Unit
Operating Average Current	TX	-	24.6	-	mA
Sleep Current		-	3.8	-	mA

Voltage Ratings

Feature	Minimum	Typical	Maximum	Unit
VCC	2.0	3.3	3.6	V

Mechanical Characteristics

Board Dimensions

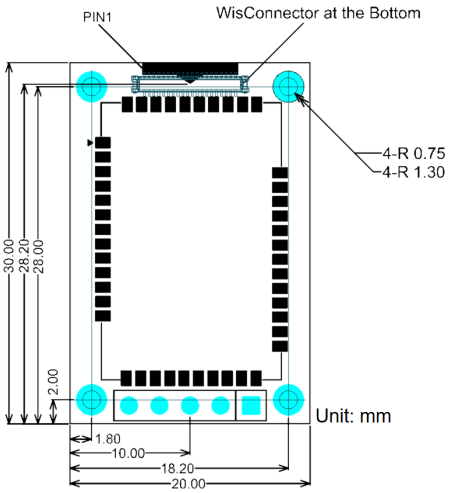
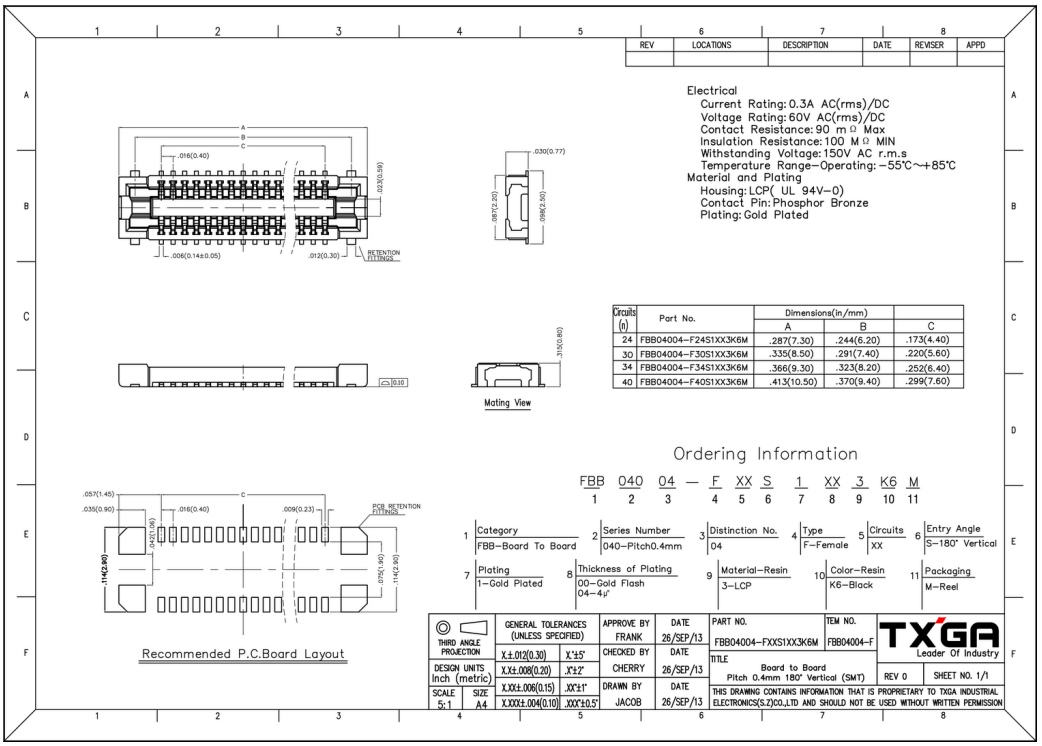


Figure 3: Mechanical Dimensions

WisConnector PCB Layout



Circuits (n)	Part No.	Dimensions(in/mm) A	B	C
24	FBB04004−F24S1XX3K6M	.287(7.30)	.244(6.20)	.173(4.40)
30	FBB04004−F30S1XX3K6M	.335(8.50)	.291(7.40)	.220(5.60)
34	FBB04004−F34S1XX3K6M	.366(9.30)	.323(8.20)	.252(6.40)
40	FBB04004−F40S1XX3K6M	.413(10.50)	.370(9.40)	.299(7.60)

Figure 4: WisConnector PCB footprint and recommendations

Schematic Diagram

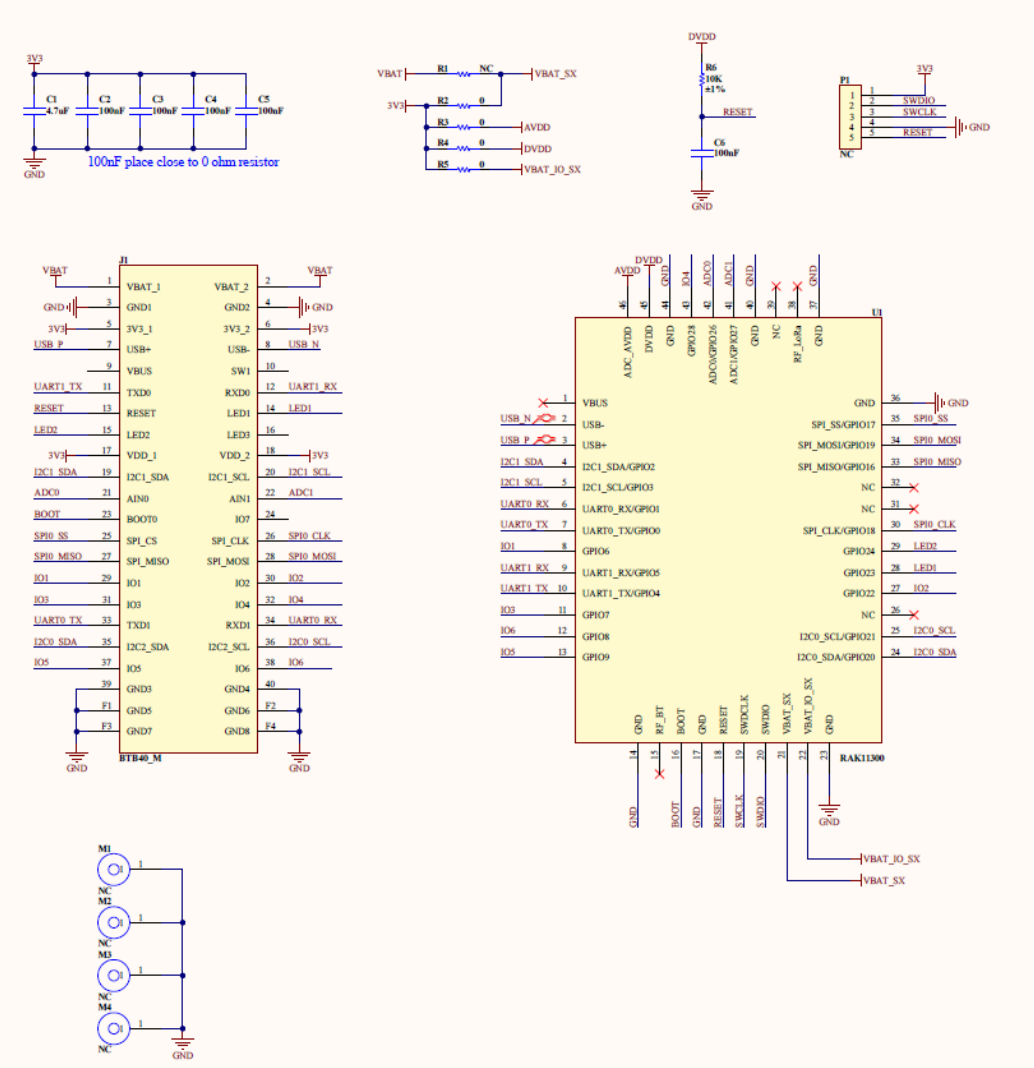


Figure 5: RAK11310 Schematic Diagram

- **WisConnector**: The breakout module exposes the RAK11310 stamp module pinout to the board-to-board WisConnector.
- **WisConnector Pin Order**: The pin order of the WisConnector is located on the bottom layer of the module.
- **Core Module**: The breakout module itself has a RAK11310 at its core, and it shows the core module pin and connection information. By default, the NFC function is disabled to preserve the low-power characteristic of this core.
- **SWD Interface**: The breakout module exposes an SWD debug interface. Additionally, the RST pin is used to reset the RAK11310 core module.
- **Power Up Automatic Reset**: The breakout module has a power-up automatic reset circuit, and the schematic shows the automatic reset mechanism. This module also can be reset through the WisBlock Base reset pin.

WisConnector Pin

Pin Number	Description	Pin Number	Description
1	VBAT_1	2	VBAT_2
3	GND1	4	GND2
5	3V3_1	6	3V3_2
7	USB+	8	USB-
9	VBUS	10	SW1

Pin Number	Description	Pin Number	Description
11	TXD0	12	RXD0
13	RESET	14	LED1
15	LED2	16	LED3
17	VDD_1	18	VDD_2
19	I2C1_SDA	20	I2C1_SCL
21	AIN0	22	AIN1
23	BOOT0	24	IO7
25	SPI_CS	26	SPI_CLK
27	SPI_MISO	28	SPI_MOSI
29	IO1	30	IO2
31	IO3	32	IO4
33	TXD1	34	RXD1
35	I2C2_SDA	36	I2C2_SCL
37	IO5	38	IO6
39	GND3	40	GND4
F1	GND5	F2	GND6
F3	GND7	F4	GND8

RAK11310 Module Pinout

Pin Number	Description	Pin Number	Description
1	VBUS	2	USB-
3	USB+	4	I2C1_SDA/GPIO2
5	I2C1_SCL/GPIO3	6	UART0_RX/GPIO1
7	UART0_TX/GPIO0	8	GPIO6
9	UART1_RX/GPIO5	10	UART1_TX/GPIO4
11	GPIO7	12	GPIO8
13	GPIO9	14	GND
15	RF_BT	16	BOOT
17	GND	18	RESET

Pin Number	Description	Pin Number	Description
19	SWD_CLK	20	SWDIO
21	VBAT_SX	22	VBAT_IO_SX
23	GND	24	I2C0_SDA/GPIO20
25	I2C0_SCL/GPIO21	26	NC
27	GPIO22	28	GPIO23
29	GPIO24	30	SPI_CLK/GPIO18
31	NC	32	NC
33	SPI_MISO/GPIO16	34	SPI_MOSI/GPIO19
35	SPI_SS/GPIO17	36	GND
37	GND	38	RF_LoRa
39	NC	40	GND
41	ADC1/GPIO27	42	ADC0/GPIO26
43	GPIO28	44	GND
45	DVDD	46	ADC_AVDD

Certification